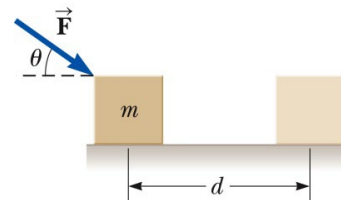


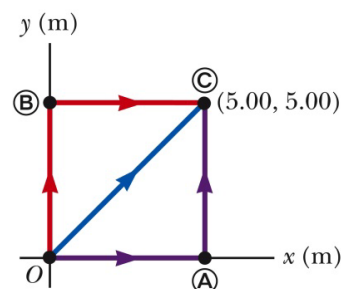
MC Problems: Electric Potential and Potential Energy

Answers to MC problems are on page M-14. **Updated Friday September 18, 2026 8:57am.**

MC1. (*Physics 131 Review Problem*) A block of mass $m = 2.5$ kg is pushed a distance $d = 2.2$ m along a frictionless, horizontal table by a constant applied force of magnitude $F = 16$ N, directed at $\theta = 25^\circ$ below the horizontal, as shown. Determine (a) the work done on the block by the applied force, (b) the work done by the normal force exerted by the table, (c) the work done by the gravitational force, and (d) the net work done on the block.

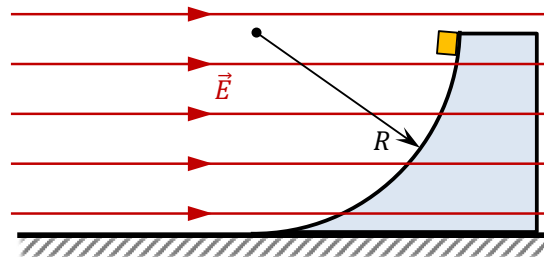


MC2. (*Physics 131 Review Problem*) A constant force $\vec{F} = (3\hat{i} + 4\hat{j})$ N acts on an object. The force does not vary with time or with the position or velocity of the object. Using the general definition for work done by a force $W = \int_i^f \vec{F} \cdot d\vec{s}$, calculate directly the work done by $\vec{F}$ on the particle as it moves from the origin O to point C along each of the three paths shown.

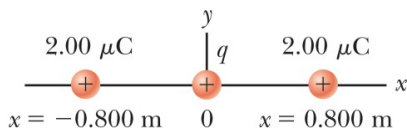


MC3. (a) Find the electric potential a distance 1.0 cm away from a proton. (b) What is the potential difference between two points that are 1.0 cm and 2.0 cm from a proton? (c) Repeat parts (a) and (b) for an electron.

MC4. A small block is released from rest at the top of a frictionless, quarter-circle ramp of radius R . The block has a mass m and a charge q , and is subject to a uniform external field $\vec{E}$ directed horizontally to the right. What is the speed of the particle when it reaches the bottom of the ramp?

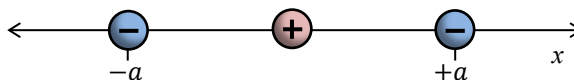


MC5. Two particles with charge $2.0 \mu\text{C}$ are placed at $x = +0.8$ m and $x = -0.8$ m. A small “test charge” with $q = 1.28 \times 10^{-18}$ C is placed at the origin. Find (a) the electric field at the origin due to the two $2 \mu\text{C}$ charges, (b) the force on the test charge, and (c) the electric potential at the origin due to the two $2 \mu\text{C}$ charges.

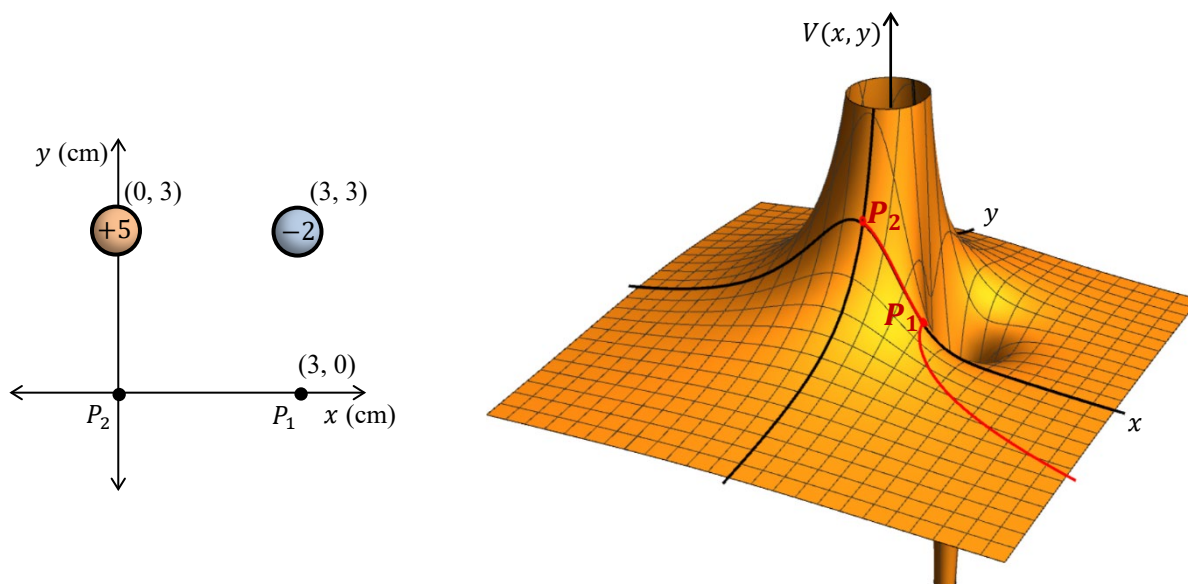


MC6. A $+4$ nC charge is at the origin, and a $+7$ nC charge is at the point $(0, -4)$ m. (a) Find the electric field at the point $(3$ m, $0)$. (b) Find the electric potential at $(3$ m, $0)$. (c) Which part was easier, part (a) or part (b), and why?

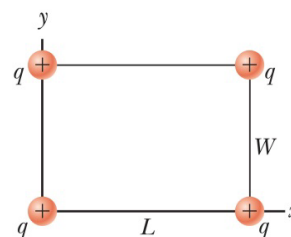
MC7. Three charged particles lie along the x axis: a charge $+Q$ is at $x = 0$, and two charges $-Q$ are at $x = \pm a$. (a) Draw a graph of potential V as a function of x along the axis. (b) Draw a graph of electric field E as a function of x along the axis, where the sign of the field indicates its direction. (c) At $x = \pm a/2$ is V positive, negative, or zero? (d) At $x = \pm a/2$ is E positive, negative, or zero? Explain your reasoning for parts (c) and (d).



MC8. A $+5$ nC charge sits on the y axis at $(0, 3$ cm), and a -2 nC charge sits at $(3$ cm, 3) as shown. (a) Find the potential V at the points P_1 and P_2 shown. (b) What is the change in potential energy ΔU of the system when a third particle, of charge $q_3 = +3$ nC and mass $m = 4$ μ g (4×10^{-9} kg), is brought in from very far away to position P_1 ? (c) How much work is required to move the $+3$ nC particle from point P_1 to point P_2 ? (d) The $+3$ nC particle is released from position P_2 , and it accelerates away from both other charges. What is the maximum velocity attained by the charge once it is very far away?



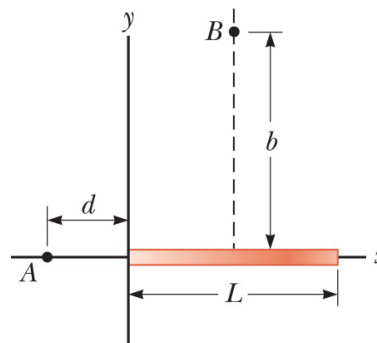
MC9. Four identical particles, all with $q = +10$ μ C are located on the corners of a rectangle ($L = 0.6$ m, $W = 0.15$ m) as shown. Calculate the change in electric potential energy of the system as the particle at the lower left corner of the figure is brought to this position from infinitely far away. (The other three particles remain fixed in position during the process.)



MC10. A thin rod of length $L = 20$ cm, with a total positive charge $Q = 10$ nC, lies on the y axis between $y = 0$ and $y = 20$ cm. In this problem, you will calculate the electric potential V at a point along the x axis a distance $x = 15$ cm from the origin. (a) First, estimate it numerically, using a spreadsheet program like Excel. Divide the rod into ten individual 1 nC charges located at $y = 1$ cm, $y = 3$ cm, etc., up to $y = 19$ cm. Treat each as a point charge, and find the potential at the point $x = 15$ cm, $y = 0$ due to all of them. (b) Now solve the problem exactly by setting up and solving an integral over the entire length of the rod.

MC11. A rod of length L lies along the x axis with its left end at the origin, as shown in the figure to the right. The rod has a nonuniform charge density $\lambda = \alpha x$, where α is a positive constant. (That is, the charge density increases linearly over the rod from left to right.) (a) What would be the standard SI units of α ? (b) Calculate the electric potential at point A in the figure.

MC12. For the arrangement described in the previous problem and pictured to the right, find the electric potential at point B , which lies on the perpendicular bisector of the rod a distance b above the axis. (For this problem, you can leave your answer in the form of an integral.)



MC13. Suppose the potential energy of a system in which a two-dimensional force acts on an object is given by $U(x, y) = 3x^3y - 7x$. (a) Find the x component F_x of the force that acts at the point (x, y) . Hint: you already know that in one dimension along the x axis, the force is related to potential energy by $F = -dU/dx$. The extra “ y ” doesn’t really change anything. (b) Reasoning by analogy,¹ write the y component F_y of the force as well. (c) What is the force $\vec{F}$ on the object, written in component form in terms of $\hat{i}$ and $\hat{j}$?

MC14. In some region of space, the electric potential is given by $V = 5x - 3x^2y + 2yz^2$, where the coordinates x, y , and z are in meters and V is in volts.² (a) Write expressions for the x, y , and z components of the electric field. (b) What is the magnitude of the field at the point $(x = 1, y = 0, z = -2)$?

MC15. An insulating rod, 14 cm long, carrying a uniform charge of $-7.50 \mu\text{C}$ is bent into a semicircle. Find the electric potential at point O , the center of the semicircle.

MC16. A particle maintains a constant velocity of 5 m/s.

- What is the change in the particle’s position Δx between the times $t = 10$ s and $t = 30$ s?
- Draw a graph of v vs. t .
- Draw several different possible graphs of position x vs. t . (The particle could start at several different initial positions, after all.)
- Write a general function for x as a function of t . Don’t forget the integration constant!
- Write a general expression for the displacement Δx , as a definite integral of $v(t)$.
- Evaluate your integral in the previous part, writing Δx in terms of v and Δt .
- Do you need an integration constant to find Δx between, say, $t = 10$ s and $t = 30$ s?



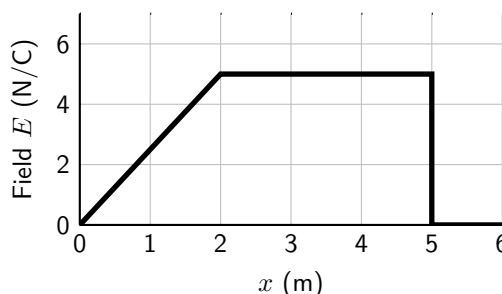
MC17. (This is a companion problem to MC16.) A uniform electric field is given by $\vec{E} = 500\hat{i}$ V/m.

- What is the change in electric potential ΔV between the points $x = 10$ cm and $x = 30$ cm?
- Draw a graph of E vs. x .
- Draw several different possible graphs of potential V vs. x . (There is no reference voltage defined, after all.)
- Write a general function for V as a function of x . Don’t forget the integration constant!
- Write a general expression for ΔV , as a definite integral of $E(x)$.
- Evaluate your integral in the previous part, writing ΔV in terms of E and Δx .
- Do you need an integration constant to find ΔV between, say, $x = 10$ cm and $x = 30$ cm?

MC18. A conducting metal sphere has a radius of R and carries a positive charge Q . (a) Draw a graph of the electric field $E(r)$ covering both inside and outside the sphere. (b) Draw a graph of the electric potential $V(r)$ both inside and outside the sphere. (c) What is the value of the electric potential at the center of the sphere, assuming the usual reference point that $V = 0$ at $r \rightarrow \infty$? (d) What can you say that is always true about the graph of electric potential inside a conductor? Hint: See workbook problem 26.7.

MC19. For the graph of electric field $E(x)$ to the right,

- sketch a graph of the potential $V(x)$, where the reference is $V(\infty) = 0$. (b) Also using $V(\infty) = 0$, write an equation for the potential $V(x)$. It’ll be different for $x < 2$ and $x > 2$, so you’ll need to use a curly brace (“}”) to separate the cases. (Be careful with your signs and the constants....)



¹That is, using the advanced mathematical procedure of “doing it for y instead of x .”

²The units have been left off of the numbers 5, 3, and 2 for visual clarity.

Answers to Selected MC Problems:

MC1. Two of them are 31.9 Joules, and two of them are zero.

MC2. At least one of them is 35 Joules.

MC3. (a) +144 nV (b) -72 nV (c) -144 nV, +72 nV

MC4. $v_f = \sqrt{2R(g - qE/m)}$. (Hint: start by drawing some equipotential lines on the figure.)

MC5. One of the three answers begins with the digits “449”.
The other two answers are zero.

MC6. (a) $(5.51\hat{i} + 2.01\hat{j})$ N/C (b) 24.6 volts

MC7. (c) $V(-\frac{a}{2}) = V(-\frac{a}{2}) < 0$ (d) $E(-\frac{a}{2}) < 0, E(+\frac{a}{2}) > 0$

MC8. (a) 460 V and 1075 V (b) 1.38 μ J (c) 1.84 μ J (d) 40.2 m/s

MC9. 8.94 J

MC10. (using $k_e = 8.99000 \times 10^9$ Nm²/C²) (a) 493.922 Volts (b) 493.826 Volts

MC11. (a) C/m² (b): $V = k_e\alpha \left[L - d \ln \left(1 + \frac{L}{d} \right) \right]$

MC12. The integral you get is $\int_0^L \frac{k_e\alpha x}{\sqrt{(x - L/2)^2 + b^2}} dx$, and as I said
in the problem, it’s fine with me if you leave it that way.

MC13. (a) $F_x = -9x^2y + 7$ (all you did here was to treat y is a constant; that is, you took a partial derivative, which means that you can look very fancy by writing it as $\partial U/\partial x$) (c) $\vec{F} = (-9x^2y + 7)\hat{i} + (-3x^3)\hat{j}$

MC14. (b) 7.07 N/C

MC15. -1.51 MV

MC16. (a) 100 m (c) Your graphs should have the same slope, but different intercepts. (d) $x = \int v dt = vt + C = vt + x_i$ (e) $\Delta x = \int_{t_i}^{t_f} v dt$ (f) $\Delta x = \int_{t_i}^{t_f} v dt = v(t_f - t_i) = v\Delta t$ (g) no integration constant required.

MC17. (a) 100 V (c) Your graphs should have the same slope, but different intercepts. (d) $V = -\int E dx = -Ex + C$ (e) $\Delta V = -\int_{x_1}^{x_2} E dx$ (f) $\Delta V = -\int_{x_1}^{x_2} E dx = -E\Delta x$ (g) no integration constant required.

MC18. (a) Remember, $E = 0$ inside a conductor. (c) $V = k_e Q/R$

MC19. $V(x) = \begin{cases} (-1.25 \text{ V/m}^2)x^2 + 20 \text{ V}, & \text{if } 0 < x < 2 \\ (-5 \text{ V/m})x + 25 \text{ V}, & \text{if } 2 < x < 5 \\ 0, & \text{if } x > 5 \end{cases}$